

ORIENTATION MODULE UNIT 1
Course Handbook

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Learning outcomes:

LO1: Recognize how to engage with and navigate the Online Campus.

LO2: Review the grade distribution, pass requirements, technical considerations, and other important course information.

This handbook provides the information you will need regarding the distribution of grades, minimum pass requirements, technical considerations, and structure of this course.

1. Course overview

This course will provide you with the skills to extract knowledge and insights from data through machine learning and advanced analytics. It begins with a foundation in data science methodologies and tools. You will then be introduced to the programming languages commonly used in data science before focusing on Python. Finally, you will explore machine learning with Python in preparation for your capstone project.

2. Learning outcomes

Learning outcomes and exit-level outcomes are the learning objectives associated with course content. Learning outcomes describe the skills that you are expected to have mastered upon completion of an activity or component in a module. They are organized to build on each other throughout a module and ultimately lead to the formation of an exit-level outcome. Each module is aligned with one or more exit-level outcomes, which describe the overall knowledge and skills you will acquire upon completion of the course.

The exit-level outcomes (ELOs) for this course are as follows:

- **ELO1:** Apply advanced data science methodologies and tools.
- **ELO2:** Investigate programming languages commonly used in data science.
- **ELO3:** Apply critical thinking and problem-solving skills to analyze complex datasets and derive actionable insights.
- **ELO4:** Create and deploy machine learning models for practical applications.
- **ELO5:** Use practical experience in executing comprehensive data science projects from start to finish.
- **ELO6:** Evaluate ethical, legal, and professional responsibilities in data science.

3. Learning times

This course consists of 10 modules, as well as the Orientation Module. Each module is released on a Wednesday at 10am (10:00) Eastern Standard Time (EST). You can consult the course calendar on the Online Campus for a full breakdown of your assignment due dates.

The core content in each module may require an average of 15 hours of your time per module. You may find that a module takes you less or more time than indicated. This is only an estimate, given that each learner may work at a different pace.

Each module is broken down into manageable chunks or units. It is recommended that you work through the components within these units in the order they are presented. A unit represents a meaningful portion of work that contains one or two key concepts that you can work through and bring into immediate application before progressing. To learn effectively, it is recommended that you complete a unit in one sitting.

The average learning time per module includes weekly live sessions. Each module has a 90-minute live session that includes 60 minutes of live instruction and 30 minutes of office hours where you can ask questions about the content.

Note:

Should you have any additional questions regarding the structure of the course, the assessments, or the grade distribution, please do not hesitate to contact your Tutor in the discussions (but do so during the week, as Tutors are not available to attend to queries over weekends).

Feel free to use Xpert, our AI assistant, which is available 24/7.

4. Participation requirements

You will need to achieve an overall grade of 70% across all activities to pass this course.

Table 1: Grading system for this course.

90%–100%	Proficient
70%–89%	Approaching proficiency
60%–69%	Partial proficiency
50%–59%	Incomplete
Below 50%	No submission

Your grade will be determined based on your performance in the following activities:

- Assessment quizzes
- Capstone project

Practice quizzes, activities, discussions, and enrichment activities are not graded. Live sessions are optional and attendance does not impact your grade.

A certificate will be issued in your legal name upon successfully completing the course according to the stipulated requirements. No certificate will be issued to you if you do not meet the stipulated requirements for the award of a certificate.

5. Assessment

The following list indicates how graded assessments are distributed throughout the course:

- **Module 1:** Assessment quiz (3.33%)
- **Module 2:** Assessment quiz (3.33%)
- **Module 3:** Assessment quiz (3.33%)
- **Module 4:** Assessment quiz (3.33%)
- **Module 5:** Assessment quiz (3.33%)
- **Module 6:** Assessment quiz (3.33%)
- **Module 7:** Assessment quiz (3.34%)
- **Module 8:** Assessment quiz (3.34%)
- **Module 9:** Assessment quiz (3.34%)
- **Module 10:** Capstone project (70%)

By the end of this course, you will have completed a capstone project that counts 70% toward your final grade. This capstone project provides you with a concrete takeaway that you can use in the real world. For the capstone project, you will utilize the new skills you have learned and create a PowerPoint presentation based on the analyses conducted in the previous modules. The capstone project modules are accessible from the start of the course if you would like to review them in more detail. Submission of the capstone project is only required on the indicated dates.

6. Technical considerations

You will need the following software to make the most of this course:

- Microsoft Word
- Microsoft PowerPoint (or similar presentation software)
- Adobe Acrobat (or similar PDF reader)

For any technical queries or difficulties, please contact your Success Advisor.

6.1 Accessing the Online Campus

The Online Campus is your online learning environment where you will access your learning content and engage in collaborative learning activities.

The Online Campus is accessible at the following URL:

<https://ibm.onlinecampus.getsmarter.com/?redirect=0>

You can also access the Online Campus from the GetSmarter mobile app. You can download the app from the [Apple App Store](#) or [Google Play](#) should you wish to make use of it during your learning journey.

Note:

When engaging with course materials on the Online Campus, you will have the opportunity to navigate between activities within a module, and at times, between pages within an activity.

To move from one activity to another within a module:

- Locate the “Previous activity” and “Next activity” buttons at the bottom of your screen and click on which activity you wish to navigate to.

To navigate from one page to the next page within a particular activity:

- Locate the “Table of contents” in the side bar on the right side of the landing page and click on the title of your choice to navigate to that page.
- Alternatively, locate the “Previous page” and “Next page” buttons toward the top (or bottom) of your screen and click on the page you wish to navigate to.
- You can also locate the activity-specific buttons at the bottom of your screen (in the case of nonlinear activities) and click on the page you wish to navigate to.

6.2 Protecting your login details

Do not divulge your login and password details to anyone. Additionally, do not leave your computer unattended when you are securely logged in on the Online Campus.